TELECOM EGYPT

INTELLIGENT ASSISTANT

Overview

Build a production-ready intelligent assistant for Telecom Egypt that transforms customer service interactions (voice, text, documents) into structured outputs and actionable insights. Handle real-world conditions including noisy recordings and Egyptian dialect variations, delivering bilingual outputs (Arabic and English).

Business Scenario & Requirements

Telecom Egypt receives thousands of customer inquiries daily. The company wants to deploy an AI-powered assistant that can:

- Answer customer questions using the Telecom Egypt official website resources
- Handle queries in Arabic, English, and Egyptian dialect - via voice or text
- Detect whether input is voice or text and respond accordingly
- Speak the answer back when voice-based; always display text alongside voice response
- Allow users to upload and query documents (PDF, DOCX, TXT, Images), uploaded content will be embedded and included in the retrieval/search context for response generation.
- Ground all responses in Telecom Egypt resources with source citations
- Provide source citations for all answers
- Chat history displaying
- Audio playback for voice responses

Knowledge Sources

Source	Description
te.eg website	Official Telecom Egypt website — primary knowledge base
User Documents	PDF, DOCX, TXT, Image formats uploaded by user
Voice Input	Live speech captured via microphone

Interaction Modes

Input Mode	Scenario	Response
Voice input	User speaks in Arabic, English, or Egyptian dialect. ASR converts speech to text, passed to RAG pipeline.	Synthesized voice (TTS) + text in chat
Text input	User types a question in the chat interface.	Textual response including relevant citations.

Required Deliverables

Deliverable	Description
Working Interface	Streamlit or Gradio app with voice input, text chat, document upload, and source citations
Working Solution	End-to-end pipeline for on-premises execution
Python Notebook	Complete code with explanations covering ASR and RAG pipeline
Presentation	PPT slides explaining system architecture, workflow, and on-premises deployment
Documentation	README with setup instructions including how to run locally

Notes

1. This case study evaluates practical RAG and ASR implementation capabilities through a Proof of Concept (POC).
2. Candidates are expected to demonstrate their technical approach, problem-solving abilities, system design thinking, and on-premises deployment considerations during the presentation.
3. External APIs may be used for benchmarking or comparison purposes, but not as the core solution.
4. The requested scope represents the baseline requirements; candidates are encouraged to showcase creativity and propose additional value-added features where applicable.

Timeline

7 days from assignment